ANES 2024 CDF Data Error Report: Identical Liberal/Conservative Thermometers

Date: January 27, 2025 **Dataset**: ANES Time Series Cumulative Data File (anes_timeseries_cdf_csv_20251211.csv) **Affected Variables**: VCF0211 (Liberal Thermometer), VCF0212 (Conservative Thermometer) **Affected Year**: 2024 only

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Summary

In the 2024 wave of the ANES Cumulative Data File, **VCF0211 (feeling thermometer: liberals) and VCF0212 (feeling thermometer: conservatives) contain identical values for all 5,521 respondents**. This appears to be a data processing error, as these variables show the expected negative correlation in all prior years.

Evidence

1. Correlation Between Variables by Year

Year	Correlation (VCF0211 vs VCF0212)	N
2008	-0.18	2,322
2012	-0.22	5,914
2016	-0.23	4,270
2020	-0.16	8,280
2024	**1.00**	**5,521**

The 2024 correlation of exactly 1.0 indicates the variables are identical.

2. Identical Values Check

```
Number of rows where VCF0211 == VCF0212 in 2024: 5,521 out of 5,521 (100%)
```

3. Mean Values by Year

Year	Liberal Therm Mean	Conservative Therm Mean
2008	49.2	52.1
2012	47.8	51.3
2016	46.9	49.8
2020	56.0	59.2
2024	**54.5**	**54.5**

The identical means in 2024 (54.5) confirm the variables contain the same data.

Impact on Research

This error affects any analysis using ideological group feeling thermometers in the 2024 ANES, including:

1. **Affective polarization measures** that include |liberal therm - conservative therm| as a component
2. **Ideological sorting** analyses comparing warmth toward ideological in-group vs out-group
3. **Thermometer difference scores** used as measures of ideological affect

Example: Affective Polarization Bias

A standard 3-component affective polarization measure (mean of party, candidate, and ideological thermometer differences) would show:

Year	3-Component Measure	2-Component Measure (excluding ideo)
2016	42.9	47.2
2020	51.2	57.0
2024	41.0	**58.4**

The 2024 3-component measure (41.0) is biased downward because the ideological difference = 0 for all respondents. The 2-component measure (58.4) shows the true pattern of continued increase.

Reproduction Code

```
import pandas as pd
import numpy as np

# Load data
df = pd.read_csv('anes_timeseries_cdf_csv_20251211.csv', low_memory=False)
df['year'] = pd.to_numeric(df['VCF0004'], errors='coerce')

# Check correlation by year
for year in [2008, 2012, 2016, 2020, 2024]:
    subset = df[df['year'] == year]
    lib = pd.to_numeric(subset['VCF0211'], errors='coerce')
    con = pd.to_numeric(subset['VCF0212'], errors='coerce')

    # Filter valid responses (0-97 range)
    valid_mask = (lib <= 97) & (con <= 97)
    lib_valid = lib[valid_mask]
    con_valid = con[valid_mask]

    corr = lib_valid.corr(con_valid)
    identical = (lib_valid == con_valid).sum()

    print(f"{year}: corr={corr:.3f}, identical={identical}/{len(lib_valid)}, "
          f"lib_mean={lib_valid.mean():.1f}, con_mean={con_valid.mean():.1f}")
```

Output

```
2008: corr=-0.180, identical=297/2322, lib_mean=49.2, con_mean=52.1
2012: corr=-0.218, identical=698/5914, lib_mean=47.8, con_mean=51.3
2016: corr=-0.230, identical=475/4270, lib_mean=46.9, con_mean=49.8
2020: corr=-0.161, identical=962/8280, lib_mean=56.0, con_mean=59.2
2024: corr=1.000, identical=5521/5521, lib_mean=54.5, con_mean=54.5
```

Partisan Analysis Code

```python
# Identify which variable is the source by examining partisan patterns
# VCF0301: Party ID (1-3 = Democrat, 5-7 = Republican)

df['pid7'] = pd.to_numeric(df['VCF0301'], errors='coerce')
df['party'] = np.where(df['pid7'].isin([1,2,3]), 'Democrat',
              np.where(df['pid7'].isin([5,6,7]), 'Republican', 'Independent'))

for year in [2016, 2020, 2024]:
    subset = df[df['year'] == year]
    for party in ['Democrat', 'Republican']:
        party_data = subset[subset['party'] == party]
        lib_mean = party_data['lib_therm'].mean()
        con_mean = party_data['con_therm'].mean()
        print(f"{year} {party}: Liberal FT = {lib_mean:.1f}, Conservative FT = {con_mean:.1f}")
```

Partisan Analysis Output

```
2016 Democrat: Liberal FT = 67.0, Conservative FT = 43.9
2016 Republican: Liberal FT = 34.5, Conservative FT = 70.7
2020 Democrat: Liberal FT = 69.5, Conservative FT = 38.3
2020 Republican: Liberal FT = 29.0, Conservative FT = 72.4
2024 Democrat: Liberal FT = 65.7, Conservative FT = 65.7
2024 Republican: Liberal FT = 29.7, Conservative FT = 29.7
```

The 2024 pattern matches the liberal thermometer values from prior years, confirming that VCF0212 was overwritten with VCF0211 data.

Identifying the Source Variable

To determine which variable contains the real data and which was overwritten, we can examine the thermometer ratings by partisan self-identification. Democrats should rate liberals higher than conservatives, and Republicans should show the opposite pattern.

Thermometer Means by Party ID

Year	Party	Liberal FT (VCF0211)	Conservative FT (VCF0212)	Pattern
2016	Democrat	67.0	43.9	Liberal > Conservative ✓

Year	Party	Liberal FT (VCF0211)	Conservative FT (VCF0212)	Pattern
2016	Republican	34.5	70.7	Conservative > Liberal ✓
2020	Democrat	69.5	38.3	Liberal > Conservative ✓
2020	Republican	29.0	72.4	Conservative > Liberal ✓
2024	**Democrat**	**65.7**	**65.7**	**IDENTICAL**
2024	**Republican**	**29.7**	**29.7**	**IDENTICAL**

Diagnosis

The 2024 values follow the **liberal thermometer pattern**: - Democrats rate both at 65.7 (similar to 2020 liberal rating of 69.5) - Republicans rate both at 29.7 (similar to 2020 liberal rating of 29.0)

If VCF0212 contained the real conservative data, we would expect: - Democrats to rate it around ~38 (as in 2020) - Republicans to rate it around ~72 (as in 2020)

Conclusion: VCF0212 (Conservative thermometer) was overwritten with VCF0211 (Liberal thermometer) values in 2024.

Likely Cause

The error pattern suggests that during CDF construction, VCF0212 was accidentally populated with VCF0211 values for the 2024 wave only. This could have occurred during:

1. Variable merging from the 2024 Time Series to the CDF
2. A copy-paste error in the data processing pipeline
3. Incorrect variable mapping during CDF update

Additional Verification

After Claude found this problem, I opened up the CSV and, yep, these columns are identical for 2024.

Contact

Discovered by: Kevin Munger (but really Claude) **Report date**: January 27, 2025